FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

FULL TEST – II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 246**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 07)** contains 7 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (08 – 11) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

   Section-A (12 – 16) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

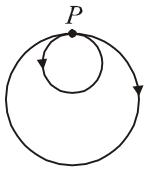
SECTION – A

(Straight Objective Type)

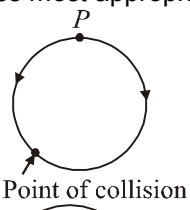
This section contains **7 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. A neutral particle at rest in a uniform magnetic field, decays into two charged particles of different masses at point P as shown in the figure. The energy released goes to their kinetic energy and particles move in the plane of the paper. Magnetic field is into the plane of paper. Select the diagram which describes path followed by the particles most appropriately.

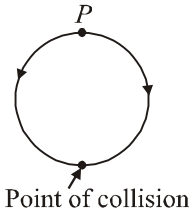
(A)



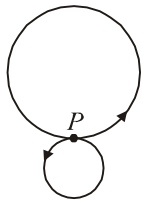
(B)



(C)



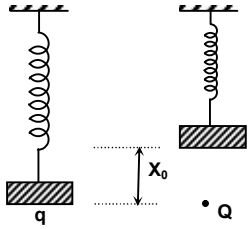
(D)



2. Choose the correct statement about centre of mass.
   (A) For calculation of any quantity associated with extended body we can assume total mass of body to be concentrated at centre of mass of the body.
   (B) Centre of mass of a body moves in the same way as if all external forces are acting on it and total mass of body is concentrated at it.
   (C) Torque of centrifugal force about centre of mass on rigid body undergoing rotational motion is always zero.
   (D) All above

Space for Rough work

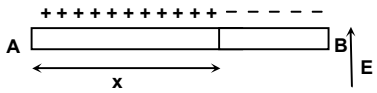
3. A block of mass m carrying charge q is suspended through non conducting spring in equilibrium spring elongation is x_0. Now another charge is moves very slowly from infinity until it is in the original position of suspended block. In this process spring acquire its natural length. Work done by external agent.



(A) $\frac{3}{2}(mgx_0)$ (B) $\frac{1}{2}(mgx_0)$

(C) $\frac{5}{2}(mgx_0)$ (D) (mgx_0)

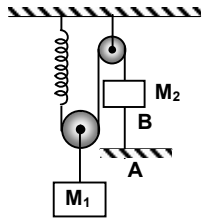
4. Rod of mass m and length ℓ carrying one charge in length x and negative charge remaining portion. Linear charge density λ is same for both charges. Rod is hinged at end A. A uniform E perpendicular to rod is switched on and rod remain in equilibrium. Find value of x. (consider gravity g is absent)



(A) $x = \frac{\ell}{2\sqrt{2}}$ (B) $x = \frac{\ell}{\sqrt{2}}$

(C) $x = \ell$ (D) $x = \frac{3\ell}{\sqrt{2}}$

5. Initially spring is relaxed and $m_1$is released from rest. When m_1 come to rest for a moment string AB is cut at the same amount. Find the acceleration of m_2 just after cutting thread.

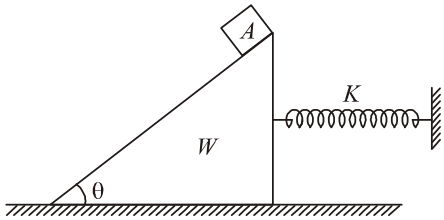


(A) $a = \left[\frac{m_1}{m_2} - 1\right]g$ (B) $a = \left[\frac{2m_1}{m_2} - 1\right]g$

(C) $a = \left[\frac{m_1}{2m_2} - 1\right]g$ (D) $a = \left[\frac{3m_1}{m_2} - 1\right]g$

Space for Rough work

6. In the adjacent figure, a block 'A' of mass m is kept on a wedge 'W' of mass 'M', which is lying at rest on a smooth horizontal surface and is connected to a rigid support through a light spring of spring constant 'K'. Initially block and wedge are at rest and spring is at relaxed state. The system (block + wedge) is set free. Calculate maximum possible compression in the spring. All surfaces are frictionless and inclination of wedge is θ.



(A) $\frac{m^2g}{K(M+m\sin^2\theta)}$ (B) $\frac{2Mmg}{K(M+m\sin^2\theta)}$

(C) $\frac{mg\sin2\theta}{K}$ (D) $\frac{2mg\cos\theta}{K}$

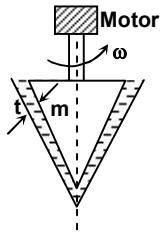
7. If an electric dipole is rotating about its centre with uniform angular velocity in anticlockwise direction in a uniform magnetic field which is in the direction of angular velocity then :
(A) Net magnetic force as well as torque due to the magnetic field on the dipole is zero
(B) Net magnetic force as well as torque due to the magnetic field on the dipole is non-zero
(C) Net magnetic force on dipole is zero but net torque due to the magnetic field on dipole is non-zero
(D) Net magnetic force on dipole is not zero but net torque due to the magnetic field on dipole is zero.

Space for Rough work

(Multiple Correct Choice Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

8. The shown figure, there is a conical shaft rotating on a bearing of very small clearance t. The space between the conical shaft and the bearing, is filled with a viscous fluid having coefficient of viscosity η. The shaft is having radius R and height h. If the external torque applied by the motor is τ and the power delivered by the motor is P working in 100% efficiency to rotate the shaft with constant ω. Then



(A) $P = \dfrac{\pi\omega^2\eta R^3\sqrt{R^2+h^2}}{2t}$

(B) $\tau = \dfrac{\pi\omega\eta R^3\sqrt{R^2+h^2}}{2t}$

(C) $P = \dfrac{\pi\omega^2\eta R^3 h}{2t}$

(D) $\tau = \dfrac{\pi\omega\eta R^3 h^2}{2t\sqrt{R^2+h^2}}$

9. A metal wire of length L, area of cross section A and Young modulus Y is stretched by a variable force F such that F is always slightly greater than the elastic forces of resistance in the wire. When the longation of the wire is ℓ, then choose the correct option(s).

(A) the work done by F is $\dfrac{1}{2}\dfrac{YA}{L}\ell^2$

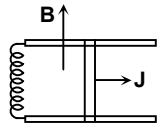
(B) the elastic potential energy stored in the wire is $\dfrac{1}{2}\dfrac{YA}{L}\ell^2$

(C) no heat is produced during the elongation

(D) the work done by F is $\dfrac{YA}{L}\ell^2$

Space for Rough work

10. Two parallel resistance less rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass m is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the correct option(s).



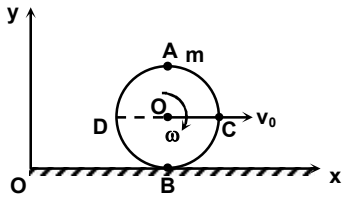
(A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{4B^2\ell^2m}}$.

(B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{4Lm}}$

(C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{B^2\ell^2m}}$

(D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{mL}}$

11. A particle of mass 'm' is attached to the rim of a uniform disc of mass 'm' and radius R. The disc is rolling without slipping on a stationary horizontal surface, as shown in the figure. At a particular instant, the particle is at the top most position and centre of the disc has speed v_0 and its angular speed is ω. Choose the correct regarding the motion of the system (disc + particle) at that instant.



(A) $v_0 = \omega R$

(B) kinetic energy of the system is $\frac{11}{4}mv_0^2$

(C) speed of point mass m is less than $2v_0$

(D) $|\vec{v}_C - \vec{v}_B| = |\vec{v}_B - \vec{v}_D|$

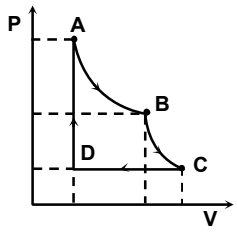
Space for Rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon one of the paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice question** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

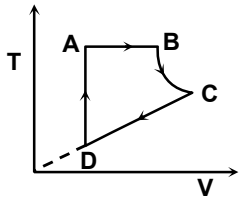
Paragraph for Questions 12 & 13

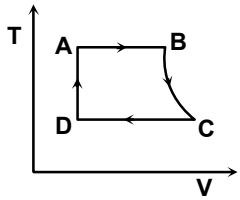
Two moles of a diatomic gas are carried through the cycle ABCDA shown in the figure PV diagram as shown in figure. The segment AB represents an isothermal expansion, the segment BC an adiabatic expansion. The pressure and temperature at A are 5 atm and 600 K respectively. The volume at B is twice that at A. The pressure at D is 1 atm.

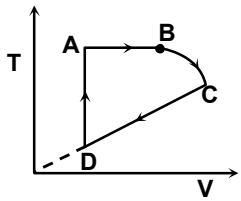


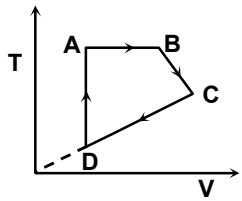
12. What is the pressure at B?
(A) 2.5 atm (B) 3 atm
(C) 3.5 atm (D) None of these

13. Find the T versus V graph of the above cyclic process.

(A)


(B)


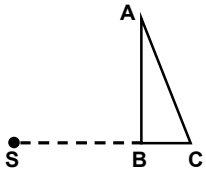
(C)


(D)


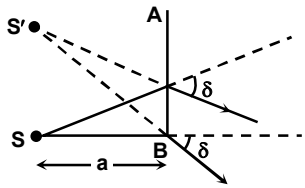
Space for Rough work

Paragraph for Questions 14 & 16

Image formation from a thin prism: For a thin prism, at near normal incidence, the angle of deviation does not vary considerably with angle of incidence and consequently we may write angle of deviation as $\delta = (\mu - 1)A$, where A is the angle of the prism and μ is the refractive index of the material of the prism. Consider a thin prism ABC and a source of light S. Suppose we want to trace the image of source S, for sake of simplicity we draw a simple ray diagram as shown in the figure.



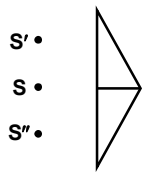
Suppose we take S as the origin, S′ as the image of S and the horizontal and vertical lines passing through S as x and y axis, and the distance SB as a (SB is horizontal). Now using geometry and using the fact that angle CAB is small, we can find the co-ordinates of S′. Now answer the following questions.



14. The x-coordinate of S′ will be
(A) zero (B) a/2
(C) 2a (D) a

15. The y-coordinate of S′ will be
(A) $(\mu - 1)Aa$ (B) $\frac{1}{2}(\mu - 1)Aa$
(C) $2(\mu - 1)Aa$ (D) $\frac{3}{2}(\mu - 1)Aa$

16. If an identical prism is joined at the base of the first prism as shown in figure and a new image S″ is formed due to it. Then the separation between S′ and S″ will be



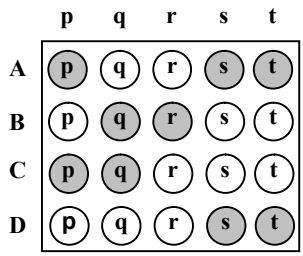
(A) $(\mu - 1)Aa$ (B) $2(\mu - 1)Aa$
(C) $\frac{(\mu - 1)Aa}{4}$ (D) $\frac{3(\mu - 1)Aa}{4}$

Space for Rough work

SECTION - B
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

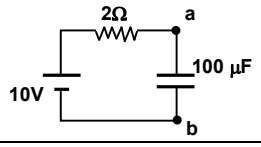
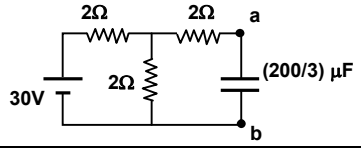
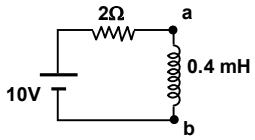
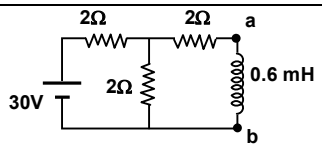


1. An electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume Bohr model to be valid.

Column I		Column II	
(A)	The electron emits an energy of 2.55 eV	(p)	$n_1 = 2, n_2 = 1$
(B)	Time period of the electron in the initial state is eight times that in the final state.	(q)	$n_1 = 4, n_2 = 2$
(C)	Speed of electron become two times	(r)	$n_1 = 5, n_2 = 3$
(D)	Radius of orbit of electron becomes 4.77A°	(s)	$n_1 = 6, n_2 = 3$
		(t)	$n_1 = 8, n_2 = 4$

Space for Rough work

2. Choose the correct equation of current in the Column- II as a function of t through the circuit element 'ab' of the circuits in the column – I

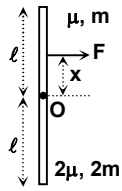
List – I		List – II
(A)		(p) $i = (5A)\left[1 - e^{-\frac{t}{(2\times10^{-4}\ sec)}}\right]$
(B)		(q) $i = (5A)\left[1 - e^{-\frac{t}{(4\times10^{-4}\ sec)}}\right]$
(C)		(r) $i = (5A)\,e^{-\frac{t}{(2\times10^{-4}\ sec)}}$
(D)		(s) $i = (5A)\,e^{-\frac{t}{(4\times10^{-4}\ sec)}}$

SECTION – C

(Integer Answer Type)

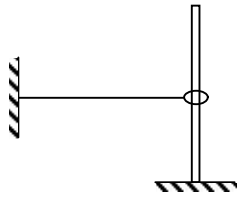
This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Two rods of same length ℓ but there masses and coefficient of friction with table are m, 2m and μ, 2μ respectively are joint at point O as shown in the figure. Find the position x from joint of two rod where force should apply perpendicular to the length of the rod to drag the rod with constant velocity is $\frac{K}{10}\ell$. Find the value of K.

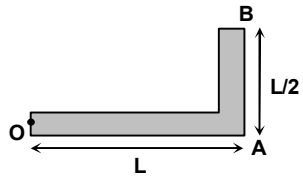


Space for Rough work

2. A string of length 0.4 m and mass 10^{-2} kg is tightly clamped at its one end and other end is free to move as shown in the figure. The tension in the string is 1.6 N identical pulses are produced at one end at equal intervals at time Δt. The minimum value of Δt which allows constructive interference between successive pulses is

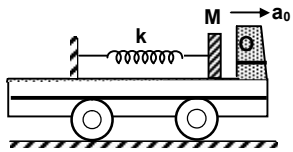


3. A uniform rod OAB is bent in L shape to form right angle at A. Length of OA is L and that of AB is $\frac{L}{2}$ respectively as shown in figure. The rod is hinged at the end O and is free to rotate in a vertical plane about O. It is set free from rest, when larger section of it is in horizontal position. Maximum angular acceleration of rod is $\frac{\sqrt{65}}{K}\frac{g}{L}$. The value of K is

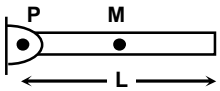


4. For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Its refractive index will be

5. A block of mass (M = 1 kg) is kept on smooth floor of a truck. One end of a spring of force constant 100 N/m^2 is attached to the block and other end is attached to the body of truck as shown in the figure. At t = 0, truck begin to move with constant acceleration 2 m/s^2. The amplitude of oscillation of block relative to the floor of truck is $\lambda/100$. Find the value of λ



6. A uniform rod of mass M is released from horizontal position as shown in the figure. The rod is free to rotate about a horizontal frictionless axis passing through hinge P. The reaction from the hinge just after the release is Mg/K. Find the value of K.



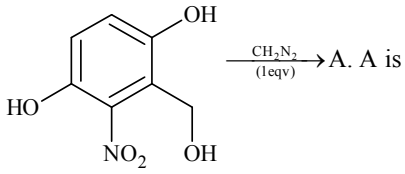
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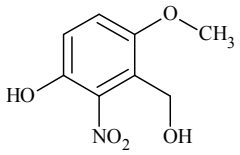
Chemistry

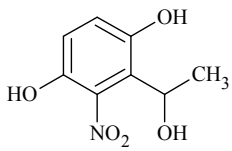
PART – II

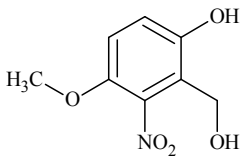
SECTION – A
Straight Objective Type

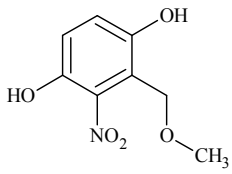
This section contains **7 multiple choice** questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONLY ONE** option is correct.

1. 

(A) 

(B) 

(C) 

(D) 

2. Choose the correct statement(s) from among the following

(A) The corrected pressure $\left(P+\frac{an^2}{v^2}\right)$ in vander waals' equation (symbols have their usual meanings) is the pressure which the gas would exert if it was ideal.

(B) Above their respective Boyle's temperature, N_2 shows throughout positive deviation ($z > 1$) while O_2 shows negative deviation ($z < 1$) followed by positive deviation ($z > 1$).

(C) A gas shows negative deviation when long range attractive intermolecular forces are dominating

(D) The intercept of PV vs. P isotherm of any gas at 27°C is equal to 24.6 L atm.

3. The ZnS (zinc blende) structure is cubic. The unit cell may be described as a face-centred sulfide ion lattice with zinc ions in the tetrahedral voids. Then among the following which is incorrect:

(A) The nearest neighbours of Zn^{2+} are four

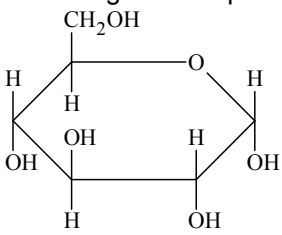
(B) The angle made by the lines connecting any Zn^{+2} to any two of its neighbours is $109^{\circ}28'$.

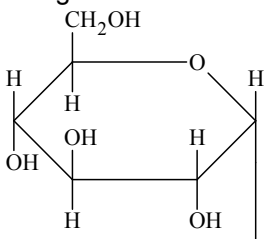
(C) The nearest neighbours of S^{-2} are four

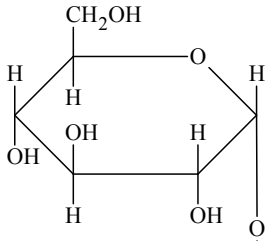
(D) The nearest neighbours of S^{-2} are eight

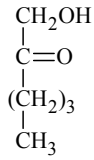
Rough Work

4. Some organic compounds are given below


I


II


III


IV

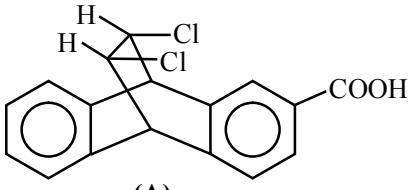
Select the correct statement about the above compounds from the following
(A) Compounds I and III would reduce Fehling's solution and will also mutarotate
(B) Compound IV will not reduce Tollen's reagent and will not mutarotate
(C) Compound II will not be reducing but mutarotating
(D) The product(s) of the acid catalysed hydrolysis of both II and III will be reducing but not mutarotating

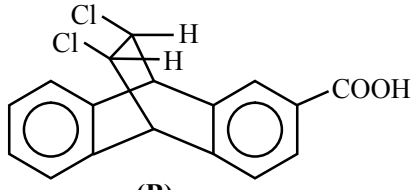
5. $Mg_3N_2 \xrightarrow{H_2O} A_{(g)} \xrightarrow[\Delta]{CuO} B + C_{(g)} + H_2O$

$C_{(g)}$ can be obtained by heating
(A) NH_4NO_3 (B) $(NH_4)_2SO_4$
(C) NH_4ClO_4 (D) $NaNO_2$ + Zn + NaOH (Conc.)

6. Which of the following statements are incorrect?
(A) Sulfate and thiosulphate ions are isostructural.
(B) Thiosulphate on reaction with HCl precipitates sulphur along with SO_2 evolution.
(C) In $S_2O_3^{2-}$ the oxidation states of the two 'S' atoms are -2 and +6.
(D) Both SO_4^{2-} and $S_2O_3^{2-}$ can decolourise I_2.

7. Difference in P^{Ka} values of compounds A and B can be best explained by


(A)
$P^{Ka} = 6.07$


(B)
$P^{Ka} = 5.67$

(A) Inductive effect (B) Electromeric effect
(C) Field effect (D) Resonance effect

Rough Work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

8. Which of the following statements are correct?
   (A) Nylon 6 and Nylon 66 are examples of condensation polymerization
   (B) D-glucose and D-galactose are epimers
   (C) Polyethene is produced via addition polymerization
   (D) Teflon has a low coefficient of friction

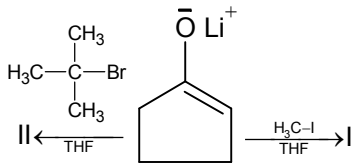
9. Consider the reaction

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} + \text{Heat}$$

What will happen if 0.5 mole of Helium gas is introduced into the vessel so that the temperature and pressure remain constant?
   (A) the equilibrium concentration of SO_2 increases
   (B) the equilibrium will shift to left
   (C) the equilibrium concentration of O_2 increases
   (D) equilibrium constant will increase

10. The standard enthalpy of formation of CO_2 will be given by :
   (A) standard enthalpy of combustion of diamond
   (B) standard enthalpy of combustion of graphite
   (C) standard enthalpy of combustion of CO
   (D) sum of standard enthalpy of formation and standard enthalpy of combustion of CO.

11. The lithium enolate base formed from cyclopentanone reacts with alkyl halides, often in different ways. As shown here, methyl iodide and *tert*-butyl bromide react to give different organic products, I and II, together with lithium halides. What are the products from these reactions?



   (A) I is 1-methoxycyclopentane
   (B) II is 1-t-butoxycyclopentane
   (C) I is 2-methylcyclopentanone
   (D) II is a mixture of cyclopentanone and 2-methylpropene

Rough Work

Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph For Question No.12 and 13

Heat of neutralization is defined as the enthalpy change when 1 mole of acid/base is completely neutralization by base/acid in dilute solution.

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l); \Delta_r H^0 = -55.84\,kJ/mol$$

$\Delta H^0_{ionization}$ of aqueous solution of strong acid and strong base is zero.

When a dilute solution of a weak acid or base is neutralized, the enthalpy of neutralization is some what less because of the absorption of heat in the ionization of the weak acid or base, for weak acid/base

$$\Delta H^0_{neutralisation} = \Delta H^0_{ionization} + \Delta_r H^0\left(H^+ + OH^- \rightarrow H_2O\right)$$

12. What is ΔH^0 for complete neutralization of strong diacidic base $A(OH)_2$ by HNO_3 ?

(A) –55.84 kJ (B) –111.68 kJ
(C) 55.84 kJ (D) 111.68 kJ

13. Under the same conditions how many mL of 0.1 M NaOH and 0.05 M H_2A (strong diprotic acid) solution should be mixed for a total volume of 100 mL produce the highest rise in temperature.

(A) 25 : 75 (B) 50 : 50
(C) 75 : 25 (D) 66.66 : 33.33

Rough Work

Paragraph For Question No.14 and 16

A green ore (A) of a metal present as a double/mixed compound is treated with HCl and then H_2S is passed in the solution. A black precipitate (B) is obtained, that is insoluble in yellow ammonium sulphide. The precipitate is dissolved in HNO_3 and then excess of NH_4OH is added. The solution becomes coloured but this colour is discharged upon addition of KCN in excess due to the formation of compound (C). The solution of (A) in H_2O liberates a colourless and odourless gas on reaction with dilute H_2SO_4 and solution of (A) gives white precipitate on addition of $BaCl_2$ solution.

14. The green ore (A) is
(A) $CuSO_4.Cu(OH)_2$ (B) $CuCO_3.Cu(OH)_2$
(C) $PbSO_4.Pb(OH)_2$ (D) $PbCO_3.Pb(OH)_2$

15. The compound (C) and its structure is
(A) $[Cu(CN)_4]^{2-}$, tetrahedral (B) $[Cu(CN)_4]^{2-}$, square planar
(C) $[Cu(CN)_4]^{3-}$, tetrahedral (D) $[Cu(CN)_4]^{3-}$, square planar

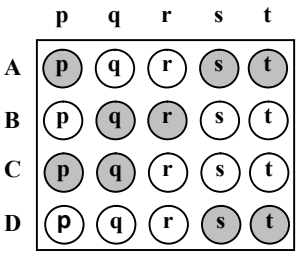
16. The procedure used to estimate the metal ion of ore (A) gravimetrically, is
(A) treatment of ore with excess KCN.
(B) treatment of solution of ore with Ag metal.
(C) treatment of ore with excess ammonia.
(D) treatment of ore with excess KSCN.

Rough Work

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. **Column – I (Compounds)** | **Column – II (Magnetic moment)**

(A) $[Fe(H_2O)_5NO]SO_4$ (p) $\sqrt{24}$ B.M.

(B) $K_2[Fe(CN)_5NO]$ (q) $\sqrt{8}$ B.M.

(C) $[Mn(CN)_6]^{3-}$ (r) Zero B.M.

(D) $[Co(SCN)_6]^{3-}$ (s) $\sqrt{15}$ B.M.

(t) $\sqrt{3}$ B.M.

2.

Column – I	**Column – II**
(A) N_2O_3	(p) Contains N – O – N linkage
(B) N_2O_4	(q) Neutral oxide
(C) N_2O_5	(r) Colored
(D) N_2O	(s) mixed Anhydride
	(t) Basic oxide

Rough Work

SECTION –C
Integer Answer Type

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

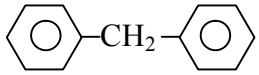
1. How many optically active isomers are possible for $CH_3CH(Br)CH(Br)CH(Br)CH_3$?

2. A certain saturated hydrocarbon effuses about half as fast as H_2O vapour. Number of carbons in the formula of this hydrocarbon are?

3. 2.505 g of hydrated dibasic acid requires 35 ml of 1N NaOH solution for complete neutralization. When 1.01 g of the hydrated acid is heated to constant weight 0.72 g of the anhydrous acid is obtained. The degree of hydration (number of water molecules) of the hydrated dibasic acid is approximately _____.

4. The number of mole of bromine required for complete bromination of 276 g of salicylic acid in aqueous medium are_____.

5. 1 g of an acid $C_6H_{10}O_4$ requires 0.768 g of KOH for complete neutralization. Determine the basicity of acid.

6. The structural formula of diphenyl methane $(C_{13}H_{12})$ is 

   How many structural isomers are possible when one of the hydrogens is replaced by a chlorine atom?

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 7 multiple choice questions numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. A gun can hit the target A with probability $\frac{27}{28}$ and target B with probability $\frac{1}{28}$, it has 47 bullets, each of which can be fired either at A or at B and each bullet may hit the target, independently of the other bullet, with probability $\frac{1}{3}$. How many bullets must be fired at A to hit the target with maximum probability
(A) 20 (B) 25
(C) 30 (D) 35

2. A polynomial P(x) of nth degree satisfies $P(k) = 2^k$ for k = 0, 1, 2,.., n, then the value of P(n + 1) is
(A) 2^n (B) 2^{n+1}
(C) $2^{n+1} - 1$ (D) 2^{n+2}

3. The number of all polynomial whose coefficients are all ± 1, that have only real roots are
(A) 6 (B) 8
(C) 10 (D) 12

4. For positive integer n, define $f(n) = 1^n + 2^{n-1} + 3^{n-2} + ... + (n-2)^3 + (n-1)^2 + n$ then the minimum of $\frac{f(n+1)}{f(n)}$ is
(A) 3 (B) $\frac{8}{3}$
(C) 2 (D) 1

Space for rough work

5. The value of $\int\sqrt{\frac{(1-\sin x)(2-\sin x)}{(1+\sin x)(2+\sin x)}}dx$ is

(A) $\ln\left(\frac{t-\sqrt{3}}{t+\sqrt{3}}\right)+2\tan^{-1}t+c$

(B) $\ln\left(\frac{t+\sqrt{3}}{t-\sqrt{3}}\right)+\tan^{-1}t+c$

(C) $\sqrt{3}\ln\left(\frac{t+\sqrt{3}}{t-\sqrt{3}}\right)+\tan^{-1}\frac{2t}{1-t^2}+c$

(D) $\sqrt{3}\ln\left(\frac{t-\sqrt{3}}{t+\sqrt{3}}\right)+2\tan^{-1}t+c$

(where $t=\sqrt{\frac{2-\sin x}{2+\sin x}}$)

6. Let x be $\left(\sqrt{108}+10\right)^{1/3}-\left(\sqrt{108}-10\right)^{1/3}$ then x is equal to

(A) 10
(B) 8
(C) 4
(D) 2

7. A circle of radius 320 units is touches internally a circle of radius 1000. The smaller circle touches a diameter of larger circle at the point p, least distance of the point p from the circumference of the larger circle is

(A) 300
(B) 360
(C) 400
(D) 420

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

8. The roots of a quadratic equation $ax^2 + bx + c = 0$ are $\alpha = z^2+z^4+.....+z^{2n}$ and $\beta = z+z^3+.....+z^{2n-1}$ where $z=e^{i\left(\frac{2\pi}{2n+1}\right)}$, $n\in z^+$ then

(A) $\alpha+\beta=1$
(B) $\alpha\cdot\beta=1$
(C) equation can be written as $x^2+x+\frac{1}{4}\sec^2\left(\frac{\pi}{2n+1}\right)=0$
(D) $\alpha+\beta=-1$

Space for rough work

9. Let $f_1(x) = x$, $f_2(x) = 1 - x$, $f_3(x) = \frac{1}{x}$, $f_4(x) = \frac{1}{1-x}$, $f_5(x) = \frac{x}{x-1}$, $f_6(x) = \frac{x-1}{x}$ suppose that $f_6(f_m(x)) = f_4(x)$ and $f_n(f_4(x)) = f_3(x)$ then

(A) m = 5 (B) n = 5

(C) m = 6 (D) n = 6

10. If a continuous function f(x) satisfy the relation $\int_0^x t f(x-t)dt = \int_0^x t\, f(t)dt + \sin x + \cos x - x - 1$, for all real number x, then which of the following does not hold good

(A) f(0) = 1 (B) f′(0) = 0

(C) f″(0) = 2 (D) $\int_0^{\pi} f(x)dx = e^{\pi}$

11. Let, $f(0, \infty) \to R$ be a strictly increasing function such that $f(x) > -\frac{1}{x}$ $\forall$ x > 0 and $f(x)f\left(f(x) + \frac{1}{x}\right) = 1$ $\forall$ x > 0, then

(A) $f(1) = \frac{1-\sqrt{5}}{2}$ (B) $f(1) = \frac{1+\sqrt{5}}{2}$

(C) $f(x) = \frac{1-\sqrt{5}}{2x}$ (D) $f(x) = \frac{(1+\sqrt{5})x}{2}$

Space for rough work

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraphs 2 multiple choice questions and based on the other paragraph 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

Read the following write up carefully and answer the following questions:
A line through the origin has the vector equation $\bar{r} = \lambda\bar{s}$, λ is a scalar parameter. Consider a family of line L with $\bar{s} = \left(\cos\theta + \sqrt{3}\right)\hat{i} + \left(\sqrt{2}\sin\theta\right)\hat{j} + \left(\cos\theta - \sqrt{3}\right)\hat{k}$ then

12. The line through origin and making acute angle $\frac{\pi}{6}$ with family of lines of the L–family are in the form of
(A) $\mu(i - j)$ (B) $\mu(i - k)$
(C) $\mu\ (i + j)$ (D) $\mu(i + k)$

13. If the lines in L meet the plane $x - z = 4\sqrt{3}$ at P. If θ varies and P describes a circle on the plane with centre on M then radius of this circle is
(A) 2 (B) $\sqrt{2}$
(C) 4 (D) $2\sqrt{2}$

Space for rough work

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:
While finding sine of a certain angle x, an absent minded professor failed to notice that his calculator was not in correct angular mode. However he was lucky to get the right answer. The two least positive value of x for which the sine of x degree is the same as sine of x radians were found by him as $\frac{m\pi}{n-\pi}$ and $\frac{p\pi}{q+\pi}$

where m, n, p, q are positive integers suppose $\frac{mn}{pq}$ be denoted by L, then

14. The value of m + n + p + q is equal to
(A) 720 (B) 900
(C) 1080 (D) 1260

15. If x measured in radian and $\lim_{x\to\infty}\left(\sqrt{Ax^2+Bx}-Cx\right)=L$, the value of $\frac{BC}{A}$ equals (A, B, C $\in$ R)
(A) 4 (B) 2
(C) $\frac{1}{2}$ (D) none of these

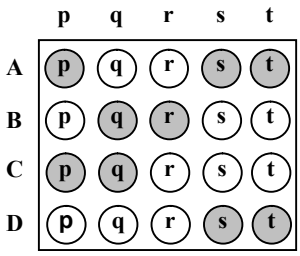
16. Assume that f is differentiable for all x, the sign of f is a follows
$f'(x) > 0$ on $(-\infty, -4)$
$f'(x) < 0$ on $(-4, 6)$
$f'(x) > 0$ on $(6, \infty)$
Let $g(x) = f(10 - 2x)$ the value of $g'(L)$ is
(A) positive (B) negative
(C) zero (D) the function g is not differentiable

Space for rough work

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following Column-I with Column-II

	Column – I	Column – II
(A)	Real value of x satisfying $\left(\sqrt{x^2-5x+6}+\sqrt{x^2-5x+4}\right)^{x/2}+\left(\sqrt{x^2-5x+6}-\sqrt{x^2-5x+4}\right)^{x/2}$ $= 2^{\frac{x+4}{4}}$ is	(p) 0
(B)	Integral value of 'a' for which the equation $(x^2 + x + 2)^2 – (a – 3)(x^2 + x + 2)(x^2 + x + 1) + (a – 4)(x^2 + x + 1)^2 = 0$ has at least one real root is	(q) 1
(C)	If P_r is the coefficient of x^r in the expansion of $(1+x)^2\left(1+\frac{x}{2}\right)^2\left(1+\frac{x}{2^2}\right)^2\left(1+\frac{x}{2^3}\right)^2$ ..... then P_1 is equal to	(r) 2
(D)	If the number of positive integral solution of x + y + z = n be denoted by P_n and $\Delta=\begin{vmatrix} P_n & P_{n+1} & P_{n+2} \\ P_{n+1} & P_{n+2} & P_{n+3} \\ P_{n+2} & P_{n+3} & P_{n+4} \end{vmatrix}$ then $\lvert\Delta\rvert$ is equal to	(s) 4
		(t) 6

2. Match the following Column-I with Column-II

	Column – I	Column – II
(A)	Let $z = (\cos 12^\circ + i \sin 12^\circ + \cos 48^\circ + i \sin 48^\circ)^6$, then Im(z) is	(p) 0
(B)	Let f(x) be a continuous and differentiable function in (a, b). If $\lim_{x\to a^+} f(x) = \infty$, $\lim_{x\to b^-} f(x) = -\infty$ and $f'(x) + (f(x))^2 = –1$ then minimum value of $\frac{b-a}{\pi}$ is	(q) 1
(C)	The number of solution of $\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x\right)=1$ is	(r) 3
(D)	The value of $\frac{1}{21}\left(\tan^2\frac{\pi}{7}+\tan^2\frac{2\pi}{7}+\tan^2\frac{3\pi}{7}\right)\left(\cot^2\frac{\pi}{7}+\cot^2\frac{2\pi}{7}+\cot^2\frac{3\pi}{7}\right)$ is	(s) 4
		(t) 5

Space for rough work

SECTION – C

(Integer Answer Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Let f(x) is a quadratic function such that f(0) = 1 and $\int \frac{f(x)dx}{x^2(x+1)^3}$ is a rational function, then the value of f′(0) is _____

2. If $0 \le [x] < 2$, $-1 \le [y] < 1$ and $1 \le [z] < 3$, (where [.] denotes the greatest integer function) then the maximum value of the determinant $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$ is _____

3. Let $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$, $\beta = \cos^{-1}\frac{4}{5}$, $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$ and $A = \frac{\cot\alpha + \cot\beta + \cot\gamma}{\cot\alpha \cdot \cot\beta \cdot \cot\gamma}$, $B = \tan\alpha \cdot \tan\beta + \tan\beta \cdot \tan\gamma + \tan\gamma \cdot \tan\alpha$ then A + B is _____

4. Let $f(x) = x^3 + 3x^2 - 6x + 11$, $g(x) = \ln|x|$. If $A = f(g'(-1))$, $B = g(f'(-1))$, $C = \int_{-2}^{1} f'(x)g'(f(x))dx$, $D = \frac{d}{dt}\int_{1}^{t^2} g(x)dx$ evaluated at t = 3 and A(B + C) – D = kln(3) then the value of k is _____

5. Let $\bar{r}_1, \bar{r}_2, ..., \bar{r}_n$ be the sides of a regular polygon inscribed in a circle of limit radius if $|\bar{r}_1 \times \bar{r}_2 + \bar{r}_2 \times \bar{r}_3 + ... + \bar{r}_n \times \bar{r}_1|$ = $|\bar{r}_1 \cdot \bar{r}_2 + \bar{r}_2 \cdot \bar{r}_3 + ... + \bar{r}_n \cdot \bar{r}_1|$ then smallest possible value of n is _____

6. A ray of light emerging from the point p(0, 4) along the tangent of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and strike a circle of radius $\sqrt{65}$ units and reflected from it. The reflected ray entering inside the ellipse through the focus (–3, 0) and strike lower surface of ellipse and gets reflected from it. If the reflected ray returns to the point p(0, 4), then ordinate of the centre of circle is _____

Space for rough work